LESSON PLAN

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Lesson Plan

Lesson Title: The motion graphs: Slope and area conversions (p-t, v-t, a-t)	**Date:** Fall 2025 **Updated Winter 2026** ***This was the lesson plan I had submitted for the course last semester with modifications applied. New knowledge from this course and others were applied to update the lesson plan to make it more effective.***
Grade Level: 11 University Preparation (SPH3U)	**Subject/Strand:** Grade 11 physics - kinematics (strand B)
Topic: Analysing and converting among position-time, velocity-time, and acceleration-time graphs in 1D	**Length of Period:** 75 minutes

Lesson Plan Description -- What do I want the learners to know and/or be able to do? What are the big ideas/enduring understandings for the lesson/unit? Indicate using 1-3 sentences.

Students learn how p-t, v-t, and a-t graphs represent one-dimensional motion and practice converting between them using slope (tangent/segment) and area. They apply this by gathering their own data in groups through a student led investigation. The goal of this lesson / lab is to investigate and apply our knowledge of motion graphs in a real world context.

STEP 1: CURRICULUM CONNECTIONS

Ontario Curriculum Overall Expectations (numbers from documents and details)	**Ontario Curriculum Specific Expectations** Numbers from documents and details -- a realistic number of expectations (1 or 2), connect to assessment.
B2. Investigate, in qualitative and quantitative terms, uniform and non-uniform linear motion, and solve related problems	**B2.2** analyse and interpret position-time, velocity-time, and acceleration-time graphs of motion in one dimension (e.g., use tangent slopes to create velocity-time graphs from position-time graphs and acceleration-time graphs from velocity-time graphs; use the area under the curve to create position-time graphs from velocity-time graphs and velocity-time graphs from acceleration-time graphs).
Learning Goals **Discuss with learners:** ***What will I be learning today?*** Clearly identify what learners are expected to know and be able to do.	**Success Criteria** **Discuss with learners:** ***How will I demonstrate what I have learned?*** Teacher/learner identify 'look fors' in language that learners can readily understand.
We are learning … - **We are learning** how to read and draw p-t, v-t, and a-t graphs for 1-D motion - **We are learning** how to convert between these graphs using slope (for velocity and acceleration) and area (for displacement)	I can … - **I can** read p–t, v–t, and a–t graphs and describe the motion with correct units and sign. - **I can** draw the matching graphs (p–t ↔ v–t ↔ a–t) using slope to link them. - **I can** find displacement from a v–t graph (area) and cross-check it on my p–t graph. - **I can** collaborate with my group by doing my role, listening, and helping my group stay on task.

STEP 2: ASSESSMENT

Indicate purpose of the assessment : **[x]** ***FOR*** **[]** ***AS*** **[ ]** ***OF***

- Most of the assessments will be focused on FOR learning (lab hand outs, chart paper group work, 3-2-1 consolidation activity).
- In the consolidation there is a part where students can reflect on their own learning which is AAL. For this lesson specifically I am focusing more on AFL. Optional: Can collect 321 sheets to assess how students are feeling in regards to the content if needed.

Indicate Achievement Chart categories being assessed
[ ] ***Knowledge and Understanding*** **[ ]** ***Thinking*** **[x]** ***Application*** **[x]** ***Communication***

Indicate Learning Skills/Work Habits:
[ ] responsibility, [ ] organization, [ ] independent work, [x] collaboration, [ ] initiative, [ ] self-regulation
(Incorporate one with Learning Goal and Success Criteria)

Indicate Assessment Mode:
Written, Oral, Performance (Write, Say, Do)

Students demonstrate learning through written work in the student booklet, graphing, calculations, poster labels, and the 3-2-1 reflection; through oral explanations during teacher conferencing, group discussion, and gallery walk feedback; and through performance by carrying out the motion investigation, collecting data, converting among p-t, v-t, and a-t graphs, and contributing to the group poster.

Indicate Assessment Strategy : What will learners do to demonstrate their learning?

Products: Booklet (graphs, calculations, answers to questions), group poster, and 3-2-1 connections

Observations: Verify in-the-moment—axes/units, best-fit line, correct conversion, correct a–t shape

Indicate Assessment Tool: Instrument used to record results/ document learning.

1) Observation Checklist (teacher-made table, 2/1/0/N)
Assesses, per student:

- **A1** Axes & units correct
- **A2** Best-fit line used on p–t
- **A3** p–t → v–t by slope (value & sign)
- **A4** v–t → a–t by slope (zero vs >0; correct sign)
- **A5** Area on v–t interpreted as displacement (and cross-checked on p–t)
- **Comm** Clarity of oral explanation
- **Collab** Contributes/rotates roles; gives peer feedback

Coding: 2 = secure · 1 = developing · 0 = with prompt · N = not observed.

NOTE: For this lesson while I have a lot on the tracking sheet that could bog me down, I can adjust and focus on fewer things on the lesson. Incorporation of a notes section on the tracking sheet can help in streamlining this assessment tracking process. Having all these different variables available on my sheet to assess can serve as reminders of what I am looking for during the lesson alongside very specific evidence I can use for later lessons.

2) Student Booklet (product) — collected in class bin
You're assessing: correct tables, labeled p–t/v–t/a–t, numerical slope with units, area calculation and match to Δx.

3) Group Poster (product)
You're assessing: alignment of the three graphs, labels/units/time window, one accurate slope statement (p–t) and one accurate area statement (v–t).

		4) **3–2–1 Connect & Reflect (product, AS)** Students will be assessing their own understanding by generalizing the steps used. As well, they will be applying what they know to check other people's work. Can put observations noted during activity in the notes columns on the tracking sheet.

STEP 3: CONSIDERATIONS FOR PLANNING

Prior Learning: What prior experiences, knowledge and skills do the learners bring with them to this learning experience?

- **Graph literacy from math.** They know how to set up and read Cartesian axes, place time on the horizontal axis, choose reasonable scales, label with units, and interpret points, segments, and piecewise changes on a graph.
- **Rate of change (slope)**. From linear functions, they can compute slope from two points, recognize horizontal lines as "zero rate," and reason about positive/negative slope indicating direction of change.
- **Area in simple shapes**. They can find areas of rectangles and triangles—needed to connect "area under a v–t graph" to displacement.
- **Units, measurement, and sign**. They are familiar with SI basics (metres, seconds), unit conversions (cm ↔ m), and using sign to indicate direction in 1-D (positive vs negative).
- **Intro kinematics language**. They've heard or used terms like position, distance vs displacement (at least informally), speed vs velocity, and acceleration (as "change in velocity"), even if distinctions need tightening.
- **Basic tech skills**. They can use an iPad/phone camera, keep it steady, and scrub slow-motion video frame-by-frame to read time stamps.
- **Position–time (p–t)**: Students have seen that the slope is velocity; a flat line means at rest; straight rising/falling lines mean constant +/– velocity; curvature indicates changing velocity.
- **Velocity–time (v–t)**: Students know the height is velocity, a horizontal line means constant velocity, crossing the time axis indicates a direction change, and area under v–t gives displacement for an interval (rectangles/triangles only).
- **Acceleration–time (a–t) baseline**: Students recognize that a = 0 corresponds to a flat v–t, and that the sign of a tells whether velocity is increasing or decreasing relative to the chosen positive direction (no calculus expected).
- **Direction & sign conventions**: Students can choose an origin and positive direction in 1-D and use signs on velocity and displacement consistently.
- **Distance vs displacement / speed vs velocity**: Students have been introduced to the difference (magnitude-only vs signed quantity) and can state which graphs show each.
- **Uniform vs non-uniform motion (qualitative)**: Students can match constant velocity to p–t (straight line), v–t (flat), a–t (zero); and constant acceleration to p–t (curved), v–t (straight, tilted), a–t (flat, non-zero).
- **From data to meaning (light numerics)**: Students can compute velocity from p–t slope using two points with units, and can check displacement by comparing v–t area to the Δx read from p–t over the same times.

IEP program implications: Accommodations, Modifications -- To be completed for Placements 2, 3 and 4

1.IEP: Written output, graphing, and assistive technology supports (Michael)

Instructional support in this lesson: clear verbal directions paired with the anchor slide and one teacher-modelled example before students begin the toy car investigation; model how to complete one row of data, one slope calculation, and one graph conversion before students work independently.

Writing/output supports: reduce unnecessary copying from the board by providing the booklet in a structured format with pre-made tables and graph spaces already included; allow point-form notes before full sentences for the booklet questions and 3-2-1 reflection; provide partially pre-labelled graph templates if needed so the focus stays on interpreting motion rather than on handwriting or setting up axes from scratch.

Assistive technology access: digital version of the booklet and consolidation sheet when helpful; allow speech-to-text, text-to-speech, or word processing for written explanations and reflections. Teachers can help with scribing / clarifying key terms and concepts as needed.

Assessment access: allow typed or oral responses during teacher check-ins; provide extra time for booklet questions, poster explanation planning, and the exit reflection; assess understanding of slope, area, sign, units, and graph relationships rather than handwriting, spelling, or mechanics.

Other notes: very social and friendly, follows directions well, and advocates for himself. He should not be placed in a role where lengthy handwriting is the only way to contribute; roles such as time-caller, data-checker, or explanation lead may support fuller participation.

2. IEP: Visual, processing, and organization supports (Anish)

Instructional supports in this lesson: provide short, chunked directions for each phase of the lesson including Minds On, mini-review, Investigation A, Investigation B, poster creation, and 3-2-1 reflection; pair all oral instructions with written instructions on the slides or board; pause between steps to check for understanding before moving on.

Organization supports: provide a supported note-taking or checklist sheet that outlines the sequence of the task, such as collect data, graph p-t, find slope, convert to v-t, convert to a-t, check area on v-t, and prepare the poster; use visual organizers for Region A and Region B so students can keep the two parts of the ramp investigation clearly separated.

Environment/grouping: preferred seating near the front during instruction and modelling; use structured group roles and visible task order to support attention and organization during the investigation.

Assessment access: allow additional processing time before responding to teacher questions; allow brief oral explanation at checkpoints before requiring a written answer; support completion of the 3-2-1 with sentence starters or guided prompts if needed.

Other notes: very social, involved, and strong in math. He is likely to do well when the visual structure of the lesson is clear and when each step is laid out in a consistent order.

3. MLL: Language and vocabulary supports (Sandra)

Instructional supports in this lesson: pre-teach and re-teach the key vocabulary needed for the lesson, such as position, velocity, acceleration, slope, area, displacement, region, transition, units, positive, and negative; pair spoken explanations with gestures, modelled graphs, and a visible word bank throughout the lesson.

Language supports: reduce reading load where possible by chunking directions and highlighting only the key task steps; provide a key vocabulary support sheet or cheat sheet; allow use of translation technology on her phone, iPad, or Chromebook throughout the lesson; use short sentence frames for booklet questions, poster explanations, and the 3-2-1 reflection. Teachers can help with scribing / clarifying key terms and concepts as needed.

Environment/grouping: place with peers she finds helpful for understanding; use clear, direct language during group directions and class discussion; ensure she can see the board, anchor slide, and model graphs clearly.

Assessment access: allow point-form responses, labelled diagrams, short sentence frames, or brief oral responses during conferencing; provide extra time; assess physics understanding and graph interpretation rather than English grammar accuracy.

Other notes: very capable with translation software and benefits from supportive peers. The visual nature of graphing can be a strength in this lesson, especially if vocabulary is reinforced consistently.

4. Additional Support: Accessible participation and motor supports (Juan)

Instructional supports in this lesson: ensure the track, meter sticks or tape marks, toy car, booklet, graph paper, and other materials are placed within reach before the activity begins; check that each part of the investigation can be completed safely and comfortably from a normal-height chair or standing position for short periods. Due to the movement involved with this activity I would recommend placing Juan and their group members in a more accessible location of the classroom so that help can be quickly provided as needed.

Physical access supports: do not require carrying, lifting, or repositioning heavier materials such as metre sticks, equipment bins, or chart paper; provide assistance for tasks requiring two-handed fine motor coordination, such as holding materials in place while writing or adjusting setup materials at the same time. Ensure that there is a chair or place that is comfortable to sit for the student nearby. Teachers can help with scribing throughout as needed or can use text-to-speech as well if easier. Consult with student prior to the lesson to gain more insight into what they need / support they would want throughout the lesson.

Environment/grouping: assign a clear role that allows full participation without unnecessary physical strain, such as videographer, timer, data analyst, or discussion lead; place with supportive peers who can assist with setup, movement of materials, or posting the group poster when needed.

Assessment access: allow oral explanation during teacher conferencing, partner-supported investigation work, and extra time for setup, transitions, and written reflection; focus assessment on observation, reasoning, graph interpretation, and communication of ideas rather than on speed of physical completion.

Other notes: gets along well with others, advocates for himself, and is hard-working and organized. He should be able to participate fully in the investigation when physical barriers are reduced ahead of time rather than addressed only after difficulty arises.

Modifications:

No curriculum modifications are planned for this lesson. All students will work toward the same learning goal of analyzing and converting among position-time, velocity-time, and acceleration-time graphs in one-dimensional motion using slope and area. The accommodations above are intended to support access to the task, not to lower or alter the expectation. Students may use different supports, response formats, or pacing, but they are still expected to engage with the same core ideas of graph interpretation, displacement, velocity, acceleration, sign, and units.

Differentiation -- How will I differentiate the instruction to ensure the inclusion of all learners? (Choose 1 or 2 areas in Year 1)

Content *(what learners are going to learn, and when):*Students learn how position-time, velocity-time, and acceleration-time graphs are connected through a lesson sequence that moves from a brief Minds On and mini-review into two linked motion investigations. Across the lesson, students revisit slope, area, sign, units, and changes in motion in visual, verbal, numerical, and hands-on ways. Key vocabulary and graph relationships are reinforced throughout so all students can access the same physics concepts.

Process *(how instruction is organized towards a gradual release of responsibility)*: Instruction follows a gradual release model. The teacher first reviews and models the key graph relationships, then guides students through the investigation setup and expectations, and then students work in structured groups to collect data, graph motion, and make graph conversions. Visual supports, group roles, teacher check-ins, prompting questions, and scaffolded tools such as templates or step lists are used as needed to support students toward greater independence.

Product *(the ways in which individual learners will demonstrate learning for assessment)*: Students demonstrate learning through multiple forms of evidence rather than one written task only. These include the individual booklet, graphing work, group poster, oral explanations during teacher conferencing, and the individual 3-2-1 reflection. Flexible response formats such as point form, typed work, labelled diagrams, and oral explanation allow students to show the same learning in accessible ways while still being assessed on the same success criteria.

Environment *(the context and environment in which learners learn and demonstrate learning)*:Learning takes place in a structured, collaborative classroom environment with visible instructions, prepared materials, and clear learning goals and success criteria. Group roles, supportive peer groupings, and teacher circulation help make participation more predictable and manageable during the investigation. Accessible materials, assistive technology, visual supports, and appropriate seating help reduce barriers so students can focus on understanding the motion graphs.

Resources and Materials & Technology Integration -- List ALL items necessary for the delivery of the lesson. Include any attachments of learner worksheets used and teacher support material that will support communication of instruction.

Minds-On:

- Tape (floor number line 0-8 m + origin) - around 10 m total
- Floor labels every 1m (paper/tape tags) - 9 labels
- Visible timer (projector / on screen) and 1 handheld timing device (backup)
- Mini whiteboards or scrap paper + clipboards - **1 per student or group**
- Dry - erase markers or pencils - **1 per student or group**

Action:

Per group (x number of groups):

- Straight “track” (2.0 m): two metre sticks or 2 m painter’s tape.
- Tick labels every 0.20 m from 0.00 to 2.00 m — 11 labels.
- Toy car with free-rolling wheels — 1.
- iPad (Slo-Mo enabled) or smartphone on airplane mode — 1 (minimum 1 per 2 groups if devices are limited).
- Stopwatch (backup) — 1 per 2 groups.
- Graph paper — 2-3 sheets per person in group.
- Ruler/straightedge — 1.
- Student Booklet (data tables + prompts) — 1 per student.
- Pencil/fine pen — 1 per student.

Class set (shared):

- 1 metre stick placed in frame for calibration during filming — 1.
- Extra painter’s/masking tape rolls — 2–3.
- Spare toy cars — 2–3.
- Device charger/power bar — 1.
- Pre-made demo axis/graph slide — 1.

Part 2 of action

Per group:

- Ramp starter: thin hardcover book or small wedge to raise the first 30–40 cm — 1.
- Optional second thin book (for “steeper ramp” Trial 4) — 1.
- Short tape strip to re-mark start line at top of ramp — 1.
- Ramp

Gallery walk:

- Chart paper - 1 sheet per group

- Colour markers (e.g., blue for part 1, green for part 2) - 2 per group
- Sticky notes - 2 per student (plus extra)
- Painter's tape for posting - 1 roll

Consolidation:

- Handout or lined paper - 1 per student
- Slide with optional sentence starters (could write on board as well)

Teacher support / assessment materials

- Full lesson slide deck
- Teacher copy of the Student Booklet / answer key
- Observation checklist / tracking sheet
- Anecdotal notes sheet
- Group role cards / role labels
- iPad / Slo-Mo quick guide for filming

Accommodation / differentiation supports

- Digital copy of the Student Booklet
- Digital copy of the consolidation handout
- Pre-labelled graph templates / scaffolded graph pages
- Written step-by-step checklist for the investigation
- Key vocabulary sheet / word bank
- Chromebook / iPad access for typing, speech-to-text, or translation as needed
- Normal-height chair / accessible workspace setup as needed

Technology / setup

- Teacher laptop / classroom computer
- Smartboard / projector
- Stable support for filming device (small stand, book, or similar)
- Charging cables / adapters
- Extension cord

Extra materials

- Extra graph paper
- Extra pencils / pens
- Erasers
- Highlighters or coloured pencils for identifying regions / transitions

Three Part Lesson

Indicate Instructional Strategy -- modelled, shared, guided, independent instruction, small group, conferencing, whole class, other _____

Modelled – brief mini-review and teacher demo of first run, axes setup, and slope calculation.
Guided instruction – teacher circulates with prompts during data collection and graph conversions.
Small groups – triads run the four trials and build graphs together.
Whole class – Minds-On "Walk the Graph," anchor slide, and debriefs.
Independent – 3–2–1 consolidation write (connections, fixes, question).
Other: Student-led inquiry lab + gallery walk (peer feedback).

Minds on: Motivational Hook/engagement /Introduction (approximately 10-20%)
How will I engage the learners (motivational strategy, hook, activation of learners prior knowledge, activities, procedures, compelling problem)?

"Walk the Graph"
Estimated time: 7-10 minutes

Purpose: This Minds-On activates prior knowledge about interpreting position–time graphs using slope and sign. Students connect real motion to simple sketches without learning any new procedures, so they feel ready for the Action where they will convert between position–time, velocity–time, and acceleration–time.

Materials: You will need painter's tape to create a floor number line from 0 m to 8 m, a stopwatch or on-screen timer that everyone can see, mini whiteboards or scrap paper for quick sketches, and markers or pencils. Having a small section of board space available to model a correct tiny graph is helpful if you need to address a common error.

Prompting Questions and Possible Learner Responses:
Questions used to activate prior knowledge and develop critical/creative thinking

NOTE: Not all prompting questions have to be used. Some prompting questions will be made up on the spot as well / not included here. Think of this more as a question bank to draw inspiration from for asking students.

What does the slope of a position–time graph represent?

Setup: Tape the number line along the front of the room, mark a clear origin at 0 m, and state that movement to the right of 0 m is the positive direction. Select one student volunteer to act as "the walker," and ensure they understand that their pace should be steady whenever asked to walk. Ask all other students to sit so they can see the number line and raise their sketches for a quick check when prompted. Additionally, I would have ideally had a random seating plan for this class so that students are not sitting alone / sitting with their friends.

Grouping: The whole class watches a single walker so everyone shares the same reference motion, while each student sketches individually on their own board or paper. This keeps the activity brisk and allows you to scan for understanding across the room in seconds. If you want peer interaction, briefly tell students to whisper-check with a neighbor before they hold up their sketch; this adds collaboration while preserving the tight timing.

Procedure (scripted flow for reference):

1. **Frame the task (30 seconds)**

Say, "You will watch our walker and, after each short motion, you will sketch one tiny position–time graph. Time goes on the x-axis and position goes on the y-axis. Work quickly, then hold your sketch up when I say boards up. Whisper-check with the person beside you before you show."

2. **Rehearse the axes and establish the origin (20 seconds)**

Point to the taped line and say, "Zero metres is here and to the right is the positive direction." Mime drawing. Say, "With your finger, show me the x-axis for time from left to right. Now show me the y-axis for position up and down."

3. **Prime attention and set timer expectation (20 seconds)**

Show the visible timer and say, "Each motion lasts five seconds. You will then have about twenty to thirty seconds to sketch. Listen for my cue boards up."

4. **Round 1: At rest, run the motion (20 seconds) and cue the sketch (20-30 seconds):**

To the walker, say, "Stand at two metres and stay still for five seconds. Ready. Go." When the timer beeps, say, "Sketch now. Make time on the x-axis and position on the y-axis. Decide what the line should do and why." After the sketch window, say, "Boards up."

5. **Check understanding, probe, and correct if needed (20-30 seconds)**

Scan the room. Say, "I am looking for a flat horizontal line since position did not change." Ask, "What does the slope tell us here" Invite a quick response such as, "The slope is zero, so the velocity is zero." If you see vertical lines, point to the axes and say, "Time always moves to the right, so rest is a flat line because position stays the same."

6. **Round 2: Constant velocity away from the origin, run the motion and cue the sketch (60 seconds)**

To the walker, say, "Start at one metre and walk steadily away from zero for five seconds. Ready. Go." When the timer beeps, say, "Sketch now. Decide the shape and the sign of the slope." After twenty to thirty seconds, say, "Boards up."

Check and narrate: "You should have a straight line with a positive slope since position increases as time increases." Ask, "What is the sign of velocity and how do you know" Invite the response, "Positive, because the slope is positive."

7. **Clarify common errors if they appear:**

If you see curves or flat lines, say, "Steady walking means constant velocity,

Possible responses: "Velocity." "How fast and in what direction the position changes."

If the position–time graph is flat, what is the motion?
Possible responses: "At rest." "Velocity is zero."

When the line slopes upward as time increases, what is the sign of velocity? How do you know?
Possible responses: "Positive velocity." "Position increases as time increases."

When the line slopes downward as time increases, what is the sign of velocity? How do you know?
Possible responses: "Negative velocity." "Position decreases as time increases."

Say the motion of Round 2 in one sentence using position, time, and velocity.
Possible responses: "As time increased, position increased at a steady rate, so velocity was constant and positive."

Say the motion of Round 3 in one sentence using position, time, and velocity.
Possible responses: "As time increased, position decreased at a steady rate, so velocity was constant and negative."

which is a straight line on a position–time graph. Curves would mean speeding up or slowing down, which we are not doing here."

8. **Round 3: Constant velocity toward the origin, run the motion and cue the sketch (60 seconds)**

To the walker, say, "Start at six metres and walk steadily toward zero for five seconds. Ready. Go." When the timer beeps, say, "Sketch now. Decide the shape and the sign of the slope." After twenty to thirty seconds, say, "Boards up."

Check and narrate: "You should have a straight line with a negative slope since position decreased as time increased." Ask, "What is the sign of velocity and how do you know?" Invite the response, "Negative, because the slope is negative."

9. **Reinforce language and peer interaction with a one-sentence share (30-40 seconds)**

Say, "Turn to your neighbour. In one sentence describe Round 3 using the words position, time, and velocity." Call one or two pairs to share.

Bridge to Action portion of lesson: Close the Minds-On by saying, "You just matched real motion to position–time graphs using slope and sign, and you explained your reasoning. Next, we will collect our own toy-car data and use our math and physics knowledge to connect position–time, velocity–time, and acceleration–time graphs."

After the activity / before the Action: Read the learning goal(s) and success criteria out loud and point to where they are posted in the room so they remain visible during the Action. Ideally these will also be posted around the classroom (on chart paper or in a visible format) for students to reference throughout the lesson.

IEP Notes and Considerations

Give directions both orally and visually, with each round posted on the board or slide, and model one simple position-time sketch before students begin so axes, units, and positive direction are clear. Michael may use a mini whiteboard, point-form notes, or a device to reduce writing demands. Anish should have the round sequence clearly visible and sit where he can easily see the number line and modelling. Sandra will benefit from simple language, key vocabulary prompts such as start, still, toward, away, positive, and negative, and peer support alongside the visual number line. Juan should be able to participate from a comfortable, accessible position without needing to move into the floor space unless he chooses to. These supports help all students access the task without changing the learning goal.

Action: During /Working on it (approximately 60%)

How will I provide practice of new concepts, and have them demonstrate new learning?

Setup before class / materials needed: Here I need the instructions and materials for the teacher if they were to do this themselves before the class starts.

Mini-review immediately after the Minds-On (3-4 minutes)

Show a single anchor slide with three reminders: the slope of a position-time graph is velocity, the slope of a velocity-time graph is acceleration, and the area under a velocity-time graph is displacement. Say, "These are the only tools we should need for today. You just need slope on a position-time in the minds on

Prompting Questions and Possible Responses:

Critical thinking questions used throughout the lesson

NOTE: Not all prompting questions have to be used. Some prompting questions will be made up on the spot as well / not included here. Think of this more as a question bank to draw inspiration from for asking students.

Mini-review anchor (immediately after Minds-On)

Q: What does the slope of a position–time graph

Overview (est. 55 minutes total)

1. Group setup, roles, and equipment check: 2 minutes (setup is down before the class for the most part)
2. Investigation A: Level track, Trials 1-2 with analysis to p-t, v-t, a-t: 20 minutes
3. Investigation B: ramp at the start into track, Trials 3-4 with analysis to p-t, v-t, a-t: 20 minutes.
4. Poster creation on chart paper and gallery walk: 8-10 minutes

Grouping, roles, and materials

Students will be working in random groups of three (methods for picking random groups include: using an online generator, numbering off students, using labeled popsicle sticks). Each group of students need one 2 m track made from meter sticks or tape marks on the floor,a toy car, visible position markers every 0.20 m (can modify depending on classroom size), a stopwatch, an ipad, graph paper, a ruler, marker or pencil, and printed student booklet. Assign roles for students at random (can do oldest to youngest by birthday for roles, or have pre-made cards that you hand to each group member at random). Another item that could be fun as well is having students rotate through each role between trials; however, it is important to note that this could take up additional time.

For the roles, one student is the videographer and uses the ipad to film the car in slow motion, a timer calls out reference time stamps to note and starts each run (can compare with ipad later), recorder marks positions, fills the table in. Students will then share the data with each other so that they can use the data together.

Ipad slow-motion quick guide

Place the iPad side-on to the track so the camera looks squarely at the car's motion, not at an angle. Keep the iPad still by resting it on a book or against a water bottle. Make sure the 0.20 m ticks are visible in the frame and that a meter stick is visible so distance can be read. Record each run in slow motion. After recording, open the video and scrub frame by frame. Pause the video when the nose of the car reaches each 0.20 m mark. Record the time shown and the corresponding position. This produces a clean table of time versus position for graphing.

Investigation part 1: Level Track (Constant Speed)

Objectives

In this first part, students will collect reliable position–time data for motion at approximately constant velocity. They will construct a clear position–time (p–t) graph with correct axes and a straight best-fit line, calculate the slope of that line, and interpret the slope as the object's velocity in metres per second. Using their own result, they will then produce a matching velocity–time (v–t) graph and an acceleration–time (a–t) graph by applying the ideas "slope of p–t is velocity" and "slope of v–t is acceleration."

Materials

Each group of three needs a 2.0 m straight "track" (two metre sticks taped end-to-end or painter's tape on a table or floor), tick labels at every 0.20 m from 0.00 m to 2.00 m, one small toy car that rolls freely, one iPad with Slo-Mo enabled (battery above 50% and storage available), one stopwatch as a backup, one Student Booklet with the data tables and graph prompts, graph paper, a ruler, and a fine pencil or pen.

represent?
A: The slope of a position–time graph represents velocity because it shows how position changes with time.

Q: If the position–time line is flat, what is the motion?
A: The object is at rest because position is not changing as time increases.

Q: If a velocity–time graph is a horizontal line above zero, what kind of motion is that?
A: That shows constant positive velocity because speed and direction are not changing.

Q: What physical quantity does the area under a velocity–time graph represent?
A: The area under a velocity–time graph represents displacement during that time interval.

Q: How do we obtain acceleration from a velocity–time graph?
A: Acceleration is the slope of the velocity–time graph because it describes how velocity changes with time.

Investigation Part 1: data capture on a level track

Q: Where exactly should you pause the slow-motion video to record a time stamp?
A: I should pause the video when the nose of the car is aligned with the tick mark so the position is precise.

Q: How can you reduce random error while timing with the iPad?
A: I can keep the iPad fixed, scrub frame by frame, and repeat the run to compare trials.

Q: If two trials give different times at the same tick, how do you decide which value to use?
A: I can average the two times or choose the cleaner value if one trial was clearly steadier.

Q: Why do we mark ticks every 0.20 m instead of irregular distances?
A: Equal spacing makes it easier to compare time intervals and spot constant or changing speed.

Q: If the car wobbles or the frame shifts, what is the best next step?
A: I should refilm that trial so the measurements are consistent and the graph is trustworthy.

Graphing and axes choices

Q: Why must time be on the horizontal axis and position on the vertical axis?
A: Time is the independent variable and position depends on it, so that placement communicates

Setup (before class)
Before students begin, place the track straight and flat and affix the 0.20 m tick labels. Mark a start line at 0.00 m and draw an arrow "+x →" to the right so the positive direction is obvious. Position the iPad side-on so the camera is perpendicular to the track and the entire 0.00–2.00 m span and the tick labels are visible. Rest the iPad on a book or against a water bottle so it remains fixed, then record a two-second Slo-Mo test clip and scrub frame-by-frame to confirm that the ticks are readable and the frame is steady. Assign and explain rotating roles: the Videographer films and keeps the device still, the Timer/Launcher counts down and releases the car at 0.00 m, and the Recorder/Grapher fills the tables and later constructs the graphs. Tell groups they will rotate roles after each trial so everyone experiences each job.

Teacher scripts and actions

1. **Introduction (2 min)**

Today we will be measuring real motion. First, we will collect data on a toy car moving at a constant speed and learn how to turn the data into a graph
Action: Model marking start line and 20 cm increments on floor.

2. **Demonstration (3 min)**

"Watch as I release the car and call out its position every 2 seconds"
Action: Release car, start stopwatch, call: "0s-0cm, 2s-40cm, 4s-80cm" call out to the nearest 20cm *could modify the distance as well (depends on classroom size which is why I went with 20 cm initially)

3. **Data Collection (8 min)**

"Now your turn. One student times, one records, and one resets the car. Repeat twice for accuracy"
Teacher circulates: Monitor correct stopwatch use, answer questions as students ask

4. **Graph Construction (8 min)**

"Now I want each group to Plot time (s) on the horizontal axis and position (cm) on the vertical axis. Label axes with units."
"Draw a best-fit line through your points, don't connect the dots exactly if they wobble."
Action: Project an example graph that you had made before: point out scales, labels, line-of-best-fit.

5. **Slope Calculation and discussion (4 min)**

"Choose two points on your best-fit line—say (2 s, 40 cm) and (6 s, 120 cm). Calculate slope: $\Delta y/\Delta x = (120-40)/(6-2)=80/4=20$ cm/s."
"That 20 cm/s is the car's speed. What would your slope look like if the car moved faster? Slower?"
Action: Write formula on board; field quick student responses.

Investigation part 2: Ramp then level (Accelerate then Constant)

Objectives
In this second part, students will observe how introducing a short acceleration phase at the start changes the shapes of p-t, v-t, and a-t graphs compared with constant-speed motion. They will identify and label the transition point where the car leaves the ramp and enters the level section, draw the correct qualitative (or semi-quantitative) shapes for v-t and a-t, explain why acceleration is positive during the ramp and near zero on the level, and use area under v-t to interpret displacement for a state time interval.

Materials
Each group reuses the same kit from Part 1. In addition, each group will

cause and effect.

Q: How do you choose a useful scale for your axes?
A: I choose scales that spread the points across most of the grid without crowding any region.

Q: Why draw a single best-fit line rather than connecting the dots?
A: A best-fit line models the overall trend and reduces the impact of small measurement errors.

Q: How do you label your axes to make the graph self-explanatory?
A: I include variable names and units, for example time in seconds on the x axis and position in metres on the y axis.

Slope and velocity on the level track

Q: How do you compute velocity from your position–time line?
A: I choose two points on the best-fit line and compute slope as change in position over change in time, which gives metres per second.

Q: Why should those two points lie on grid intersections if possible?
A: Grid intersections make the rise and run easy to read and reduce rounding error.

Q: What would your position–time graph look like if the car moved faster?
A: The line would be steeper because the position would increase more for each second.

Q: What sign would your slope have if the car moved back toward the start?
A: The slope would be negative because position would decrease as time increases.

Q: How can you check that your velocity value matches your graph visually?
A: I compare my numeric slope to the steepness of the line and see whether it matches the general rise over run.

Q: How does uncertainty in your time stamps affect the slope?
A: Small timing errors change the rise over run slightly, so repeating trials and using a best-fit line reduces that effect.

Converting p–t → v–t → a–t (Part 1)

Q: If your position–time graph is a straight line, what shape is your velocity–time graph?
A: It is a horizontal line because constant slope on

need a ramp (available in physics classrooms that I have seen)that has tape labels. Additionally you would need something to mount the ramp onto in order to create an incline (a chair, desk, books etc).

Setup
Before Part 2 begins, slide the book under the first 30–40 cm of the track so that 0.00 m sits at the top of the ramp. Confirm that the 0.20 m tick labels remain visible and that the iPad still frames the entire 0.00–2.00 m span. In the Student Booklet, have groups add a mark where Region A (ramp) ends (for example, at 0.40 m) and Region B (level) begins, so they can refer to those regions when they sketch.

Instructions (teacher script and student actions)
Step 1 — Frame what will change (about 1 minute).
Say, "Part 2 adds a short ramp so the car speeds up first and then rolls on the level at roughly constant speed. You should expect p–t to curve up during the ramp because the slope is increasing, v–t to tilt upward during the ramp and then flatten on the level, and a–t to be a flat positive line during the ramp and near zero on the level."

Step 2 — Capture Trial 3 with the ramp (about 5–6 minutes).
Have groups rotate roles. Instruct the Timer/Launcher to release the car at 0.00 m at the top of the ramp and not touch it again. The Videographer films in Slo-Mo with the device fixed. The Recorder logs the time when the nose of the car reaches each 0.20 m tick as before and circles the tick where the car clearly leaves the ramp (for example, 0.40 m). As you circulate, prompt with, "Between 0.00 m and 0.40 m, do the time gaps between ticks get smaller as you go? If they shrink, that shows the car is speeding up."

Step 3 — Confirm with Trial 4 or compare a slightly steeper ramp (about 3–4 minutes).
Ask groups either to repeat with the same ramp height to confirm the pattern or to add a second thin book to increase the angle slightly and observe a larger acceleration. Have them label the table accordingly as "Trial 4 (same ramp)" or "Trial 4 (steeper ramp)."

Step 4 — Sketch the three linked graphs with regions labelled (about 6–7 minutes).
Say, "Use new axes for Part 2. Plot enough points to show that your p–t curves up during the ramp (Region A), then becomes straight on the level (Region B). In some cases you may have more than 2 regions on your graph. For now let's try to group it into two distinct regions. Draw the v–t that matches: a line that rises during the ramp and then becomes horizontal on the level. Draw the a–t last: a flat line above zero during the ramp, then a line near zero on the level." If groups have robust timestamps, encourage them to estimate a slope on the level section to set the height of the flat segment on v–t; otherwise, accept qualitative height with correct shape and sign. As you circulate, correct two common errors verbally: "Constant speed gives a flat v–t; curves on v–t mean changing velocity," and "If v–t is flat on the level, its slope is zero, so a–t must sit at zero in that region."

Step 5 — Explain the comparison and use area on v–t (about 4–5 minutes).
Ask students to write three sentences in their booklets. First: "In Part 1 (level), p–t was straight, so v–t was flat and a–t was 0, because the slope did not change." Second: "In Part 2 (ramp → level), p–t first curved up then position–time means constant velocity.

Q: If your velocity–time graph is horizontal, what is your acceleration–time graph?
A: It is a line at zero because the slope of the velocity–time graph is zero.

Q: What does the area under your velocity–time graph from 0 to 5 seconds represent?
A: It represents the displacement over that interval and equals velocity times time for a constant velocity.

Q: If your computed velocity is 0.20 m/s, where should you draw your velocity–time line?
A: I should draw a horizontal line at 0.20 on the velocity axis for the measured time range.

Q: Why does your acceleration–time graph sit exactly at zero for the level track run?
A: It sits at zero because the velocity is not changing, so acceleration is zero.

Q: How would measurement noise show up on your three graphs?
A: It would appear as small scatter around the best-fit line on position–time, tiny height fluctuations on velocity–time, and near-zero jitter on acceleration–time.

Investigation Part 2: ramp then level

Q: What visual cue on the position–time graph tells you the car is speeding up on the ramp?
A: The curve bends upward so the slope increases with time, which indicates increasing velocity.

Q: How should the velocity–time graph reflect the ramp phase?
A: It should tilt upward because velocity increases during the ramp.

Q: What should happen to the velocity–time graph once the car reaches the level section?
A: It should become horizontal because the velocity becomes approximately constant.

Q: What should the acceleration–time graph look like during the ramp phase?
A: It should be a flat positive line because acceleration is roughly constant and greater than zero.

Q: What should the acceleration–time graph look like on the level section?
A: It should return to approximately zero because velocity stops changing.

straightened; v–t first rose then flattened; a–t was positive then near 0." Third: "Between 0–4 s on our v–t in Part 2, the area equals ___ m (show a short calculation or estimate), which represents displacement over that interval." Invite two groups to share one sentence each so the class hears the correct language.

What you should see at the end of Part 2.
Every group should have a completed ramp data table for Trials 3 and 4, a clearly labelled transition tick where the ramp ends, and three graphs with the correct shapes and region labels: p–t curving then straight, v–t rising then flat, and a–t positive then near zero. Their written explanations should reference slope, sign, and area accurately.

Gallery Walk - Poster Creation and Peer Feedback
NOTE: Depending on time, the gallery walk can be switched to just a few groups presenting their work. During the creation of the posters, the teacher can circulate to assess the posters and then ask a few exemplar (without telling the students that their work is good or bad) posters to present. This would be good as well for the consolidation to help narrow down understanding prior to the 3-2-1 final activity.

Objectives
The gallery walk allows students to communicate and compare how p–t, v–t, and a–t relate in constant-speed versus ramp-then-level motion and to justify their interpretations using precise language (slope, sign, area, displacement) and clear visuals (aligned axes and units). This activity provides students with some AAL and an opportunity to see other work. The gallery walk at the end of the action helps create a nice transition to the consolidation portion of the lesson.

Materials
Each group uses one sheet of chart paper, two colours of markers (for example, blue for Part 1 level graphs and green for Part 2 ramp graphs), several sticky notes for peer feedback, and painter's tape for posting.

Setup
Tell groups that their poster must show three aligned panels on the same sheet—p–t at the top, v–t in the middle, and a–t at the bottom—with axes and units clearly labelled and the time window indicated. Ask them to distinguish Part 1 and Part 2 with colour, and to label "Region A: ramp" and "Region B: level" on the Part 2 panels. Require two short annotations: one that reads "Slope of p–t = velocity; our Part 1 velocity ≈ ___ m/s," and one that reads "Area under v–t (0–4 s) ≈ ___ m (displacement)."

Instructions (teacher script and student actions)
Poster construction (6-8 minutes).
Say, On your chart paper include the following (from slide):

- Attach your three graphs (glue/tape): p–t (top), v–t (middle), a–t (bottom).
- Label regions on all three: "Region A: ramp" and "Region B: level," with a vertical dashed transition line
- Show slopes for each region and how you calculated them
- Compute the area under your v–t over a stated time interval and state what it represents (displacement, in metres).
- Write what the slope and area represents in our motion graphs

Q: Where on your graphs can you mark the transition from ramp to level?
A: I can mark the tick where the ramp ends on position–time, then align that time with the point where velocity–time flattens and acceleration–time drops to zero.

Q: If you make the ramp slightly steeper, what changes on the three graphs?
A: The position–time curve bends up more strongly, the velocity–time line rises more rapidly, and the acceleration–time level is higher during the ramp.

Q: How can you estimate the constant velocity on the level section from your data?
A: I can compute the slope of the straight segment on position–time or use the height of the flat segment on velocity–time.

Data quality and error analysis

Q: What is the biggest source of error in your measurements and how did you reduce it?
A: Frame selection in slow motion is the biggest source, and we reduced it by keeping the iPad fixed, scrubbing carefully, and repeating trials.

Q: How do outliers reveal themselves on your position–time graph?
A: Outliers sit far from the trend line, and if one point is far off we check the video timing or refilm.

Q: Why is a best-fit line preferable to connecting points when data are noisy?
A: A best-fit line captures the overall trend while de-emphasizing random scatter.

Q: How could you check whether friction changed between trials?
A: I could compare the slopes of the level sections across trials; a lower slope on a later trial might suggest more friction.

Q: If two groups have different velocities on the level track, what explanations make sense?
A: Differences can come from slightly different pushes, wheel friction, track alignment, or timing precision.

Gallery walk synthesis

Q: On this poster, where is velocity largest and how do you know from both graphs?
A: Velocity is largest where the velocity–time graph is highest and that matches the steepest slope on the

Posting and touring (about 2–3 minutes).
Say, “Post your chart on the wall. Visit two posters that are not yours. On a sticky note, leave one accurate connection you see between two graphs and one question you still have.”

Call-back debrief.
Point to one poster and ask, “Where is velocity largest on these graphs, and what p–t feature matches that?” (Students should point to the highest v–t value and the steepest p–t slope.) Then ask, “What does the shaded area on this v–t graph represent for that interval?” (Students should reply, “The displacement during that time.”) Close by thanking students for precise slope and area language and cue them to return materials: “Please return cars and iPads to the front table, remove the ramp books, and peel up your tape ticks.” This portion is to give students some insight into the different work displayed by students and creates a smooth transition into the consolidation portion of the lesson.

What you should see at the end of the gallery walk.
Each group should have a posted, clearly labelled three-panel poster showing Part 1 and Part 2 results in two colours with the required annotations, and each poster should have at least two peer sticky notes containing one correct connection and one thoughtful question.

IEP Notes and Considerations
Chunk and post each phase of the task in order, such as data collection, graphing, slope, graph conversion, area, and poster, so students can track the investigation clearly. Michael should have access to assistive technology, reduced copying, and scaffolded graph templates or pre-labelled axes as needed. Anish should have written instructions, a visible checklist, structured group roles, and extra processing time between steps. Sandra should have access to translation technology, a vocabulary support sheet, sentence frames, and supportive peer grouping during the booklet and poster work. Juan should have materials within reach, access to a normal-height chair, and a group role that avoids lifting or two-handed fine motor tasks. Teacher conferencing and observation may be used to collect oral evidence when written output is a barrier. No curriculum modifications are planned, as all students are still working toward the same graphing and motion-learning goal.

position–time graph.

Q: What does the shaded area on the velocity–time graph represent for this interval?
A: It represents displacement during that interval and its value equals the geometric area.

Q: Which feature shows acceleration most clearly and why?
A: The slope of the velocity–time graph shows acceleration most clearly because it indicates how quickly velocity changes.

Q: How does the sign of velocity affect displacement on the velocity–time graph?
A: Positive velocity adds positive displacement, while any portion below zero subtracts displacement because the area is negative.

Q: If two posters show the same displacement with different shapes on velocity–time, how can that happen?
A: Different shapes can enclose the same total area, so the objects can travel the same displacement with different velocity profiles.

Q: What single sentence connects all three graphs for the ramp case?
A: During the ramp the position–time slope increases, so velocity rises, which means the velocity–time slope is positive and acceleration is a constant positive value.

Consolidation & Connection (Reflect and Connect) (approximately 10 - 20%)
How will I bring all the important ideas from the learning experiences together for/with learners? How will I check for understanding?

Approx time around 10 minutes

Consolidation - “3-2-1 Connect and Reflect”

Purpose
This consolidation (not an exit ticket) verifies that students can connect p-t, v-t, and a-t using slope and area, and it prompts metacognition about what helped them learn and what they will adjust next time

Materials
Each student uses their own group’s graphs from Part 1 and Part 2, a half sheet “3-2-1” handout (or just a sheet of paper), and a pen or pencil.

Setup
Display a slide with the three writing prompts. Ask students to keep their

Prompting Questions:
Questions used to assess learners’ understanding

NOTE: Not all prompting questions have to be used. Some prompting questions will be made up on the spot as well / not included here. Think of this more as a question bank to draw inspiration from for asking students.

On your level run, what is the numerical slope of your position–time line, and what physical quantity does it represent with correct units?

graphs visible (ideally the graphs will be displayed for the gallery walk), and to write in full sentences that cite specific features on their graphs (for example, a slope value, a region label, or a time interval).

Procedure and script

1. **Frame the task**

Teacher script: "We will finish with a short 3–2–1 to connect our three graphs and to reflect on what helped you learn today. Keep your p–t, v–t, and a–t graphs visible. You will write three connection sentences, give two specific fixes to a partner, and add one question you still have."
Student action: Students place their graphs on the desk and take out a half-sheet or notebook page.

2. **Individual write - "3 connections"**

Teacher script: "Write three sentences that each connect two or three of your graphs using slope or area. Use evidence from your data, not general statements. Use these starters exactly if you like:
'On our level run, the slope of p–t was __, so v–t is __ because __.'
'During the ramp in Region A, p–t __, therefore v–t __ and a–t __ because __.'
'From __ s to __ s, the area under v–t is __, which matches the change in position on p–t from __ m to __ m.'
Work silently for two minutes."
Circulation prompts: "Point to the slope you used." "Show me the time interval that matches your area statement."

3. **Peer - "2 Fixes"**

Teacher script: "Trade papers with a partner. Read each sentence and write two clear improvements for your partner. Look for a missing unit, a missing sign, an axis label, or a mix up between height and slope. Begin your feedback with 'Fix 1' and 'Fix 2' and be specific. You have two minutes."
Circulation prompts: "Did they name the sign of velocity." "Did they state metres, seconds, or metres per second."
Student action: Partners annotate two concrete fixes on each other's page, return pages, and the writer makes the edits immediately.

4. **Whole class share - "1 Question"**

Teacher script: "Now write one question you still have about your graphs or about a poster you saw. Begin with 'I am still wondering…' or 'I am not yet sure why…' Write for thirty seconds."
Teacher script to class: "Two quick shares. What is one question you still have."
If a common misconception appears, deliver this correction: "If v–t is flat, acceleration is zero even if the v–t value is high. Height shows velocity. Slope shows acceleration."
Student action: Two or three students share one sentence each. Writers add a small star beside any sentence they corrected today.

5. **Self check and hand in**

Teacher script: "Underline the strongest connection sentence you wrote. Draw a box around the sentence you improved after partner feedback. At the bottom of the page, finish this line: 'Next time I will check ___ first.'

IEP Notes and Considerations

Keep the 3-2-1 prompts and sentence starters visible and allow flexible response formats such as full sentences, point form, typed responses, labelled diagrams, or brief oral conferencing. Michael may need to type or explain part of his reflection orally rather than handwrite extended responses. Anish may benefit from the prompts being read aloud, left visible, and supported with extra processing time. Sandra should have access to translation technology, reduced language load where needed,

At one specific time you choose, describe what each of your three graphs shows simultaneously: the point on p–t, the height on v–t, and the value on a–t. How do they agree?

In Region A (the ramp), what feature of your p–t graph proves that velocity is increasing, and how is the same idea shown on your v–t and a–t graphs?

For the interval 0–4 s, compute the area under your v–t graph and state the displacement. Does this match the change in position read from your p–t graph over the same times?

Where in your data is acceleration equal to zero, and how do two different graphs confirm that conclusion?

If the car were released to move back toward the start, how would the signs and shapes change on p–t, v–t, and a–t?

Choose two clear points on your p–t best-fit line and recompute the slope. Does the value match the height of your v–t line within reasonable error? Explain any difference.

On your v–t graph, identify an interval with constant speed. How can you prove it is constant using both v–t and p–t evidence?

Compare the ramp and level sections: where is velocity largest, where is acceleration largest, and what single sentence explains how those answers appear across all three graphs?

and support from sentence starters and key vocabulary. Juan should have enough time and a comfortable workspace for writing or conferencing. Assessment should focus on students' understanding of slope, area, sign, and graph relationships rather than handwriting, spelling, or grammar.	

Extension Activities -- What will learners do when work is completed? What will learners do if they finish early?

- Convert a curved p-t segment (qualitatively use tangent slope idea to sketch a matching v-t)
- Include negative regions on v-t and discuss how crossing v=0 shows a change in direction.
- For this can make up a graph on the board as well or use some of the prompting questions as well

Next Steps -- Where will this lesson lead to next?

- Follow-up lesson: derive and apply average-velocity and displacement formulas from a constant-acceleration v-t graph; begin B2.3 problems

My Lesson Reflection - Placement Only

This portion of the lesson will be completed following the conclusion of the lesson (what went well, what will I change - see reflection questions)

Not required for this but going to write a reflection here more so for myself

With this lesson being one of the first lessons that I had created during my first year of teachers college there is a lot I have learned since. One of the main things that I recognize after completing this lesson is that I would probably plan something less intense in the event a supply were to come on this day. As well I have found that since writing and learning more about lesson plans that I have become better at being more concise with the information I put into the plan. Furthermore, for science I find that with the help of my fellow TCs I have been able to really shift my planning approach from lessons being teacher led to more lessons being student led. With lessons being more student led (which this one is planned to be student led yay go first semester me!) I find it much easier to differentiate and to add accommodations in to further support students. The student-led lessons also allow for more natural gear shifts that keep students engaged and result in an overall more rich learning experience for the students. When it comes to the activity after being in a classroom, I may break this across two periods for the students rather than one period so that students really get the most out of each step. Or to adjust this lesson I could remove certain parts so that I can focus specifically on areas in which students may need to improve more on. It would really depend on the readiness of the students in the classroom as well as how many students there are in the class for running this lesson. I would most likely need to end the lesson prior to this one with some information on the setup and the lab so that the time is being used effectively to ensure that we get through everything. I included lots in this lesson plan as it is easier to take activities out if you run out of time than to try and incorporate effective engaging activities on the fly.